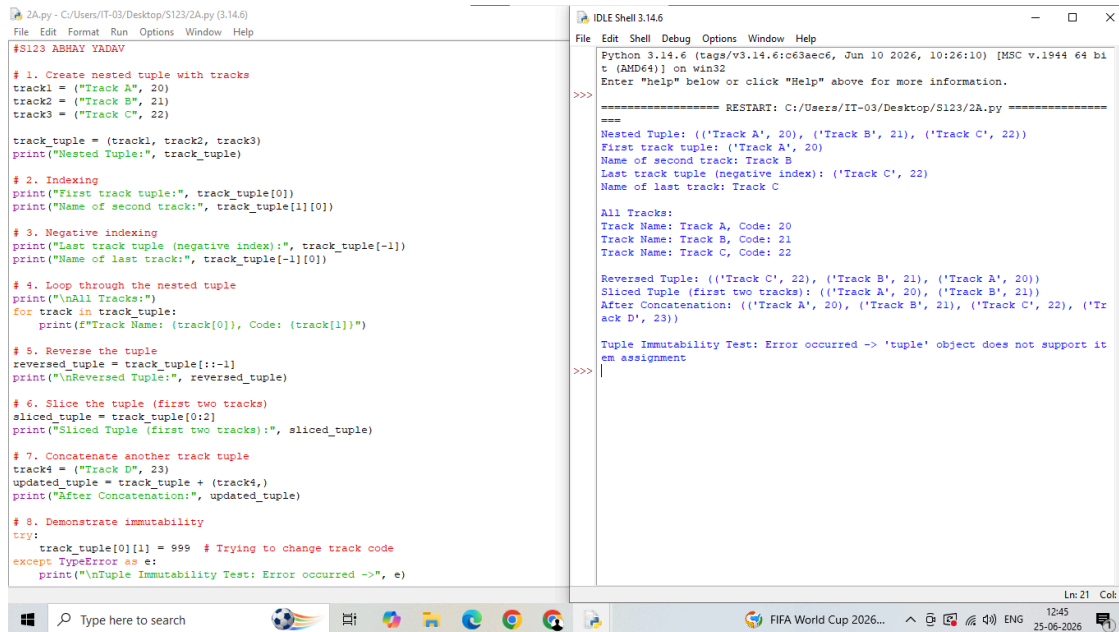


SHETH L.U.J. AND SIR M.V. COLLEGE

Subject: Python for Data Science

Practical No. 2

2a. Write a python code to create nested tuples.



```
#S123 ABHAY YADAV

# 1. Create nested tuple with tracks
track1 = ("Track A", 20)
track2 = ("Track B", 21)
track3 = ("Track C", 22)

track_tuple = (track1, track2, track3)
print("Nested Tuple:", track_tuple)

# 2. Indexing
print("First track tuple:", track_tuple[0])
print("Name of second track:", track_tuple[1][0])

# 3. Negative indexing
print("Last track tuple (negative index):", track_tuple[-1])
print("Name of last track:", track_tuple[-1][0])

# 4. Loop through the nested tuple
print("\nAll Tracks:")
for track in track_tuple:
    print(f"Track Name: {track[0]}, Code: {track[1]}")

# 5. Reverse the tuple
reversed_tuple = track_tuple[::-1]
print("\nReversed Tuple:", reversed_tuple)

# 6. Slice the tuple (first two tracks)
sliced_tuple = track_tuple[0:2]
print("Sliced Tuple (first two tracks):", sliced_tuple)

# 7. Concatenate another track tuple
track4 = ("Track D", 23)
updated_tuple = track_tuple + (track4,)
print("After Concatenation:", updated_tuple)

# 8. Demonstrate immutability
try:
    track_tuple[0][1] = 999 # Trying to change track code
except TypeError as e:
    print("\nTuple Immutability Test: Error occurred ->", e)
```

```
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

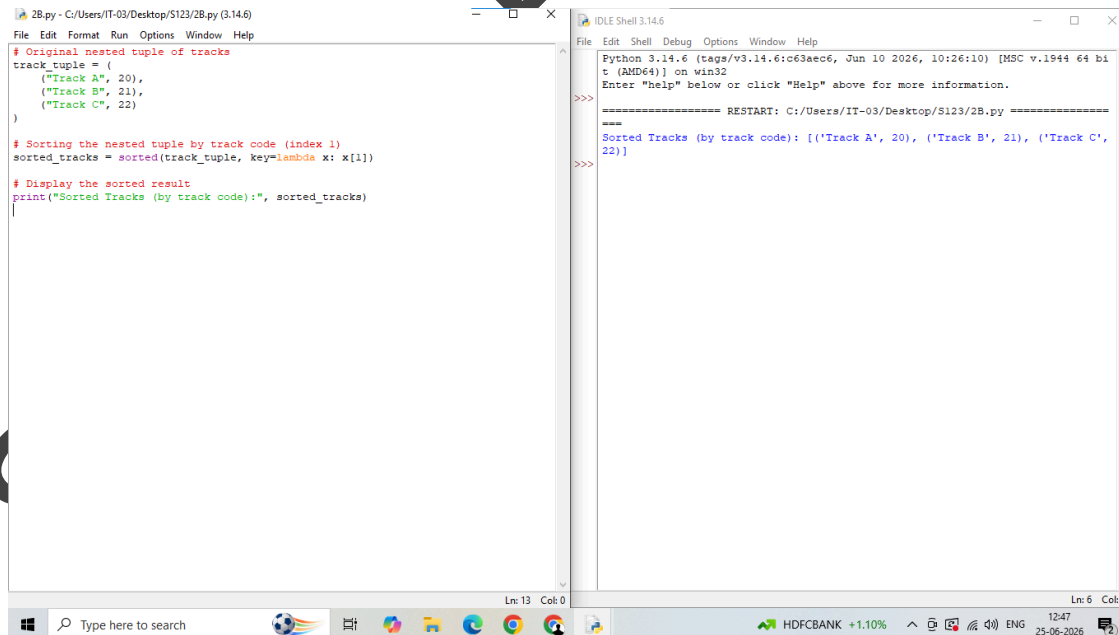
===== RESTART: C:/Users/IT-03/Desktop/S123/2A.py =====
Nested Tuple: (('Track A', 20), ('Track B', 21), ('Track C', 22))
First track tuple: ('Track A', 20)
Name of second track: Track B
Last track tuple (negative index): ('Track C', 22)
Name of last track: Track C

All Tracks:
Track Name: Track A, Code: 20
Track Name: Track B, Code: 21
Track Name: Track C, Code: 22

Reversed Tuple: (('Track C', 22), ('Track B', 21), ('Track A', 20))
Sliced Tuple (first two tracks): (('Track A', 20), ('Track B', 21))
After Concatenation: (('Track A', 20), ('Track B', 21), ('Track C', 22), ('Track D', 23))

Tuple Immutability Test: Error occurred -> 'tuple' object does not support item assignment
```

2b. Write a python code to sort the nested tuple using sorted() function.



```
# 2B.py - C:/Users/IT-03/Desktop/S123/2B.py (3.14.6)

# Original nested tuple of tracks
track_tuple = (
    ("Track A", 20),
    ("Track B", 21),
    ("Track C", 22)
)

# Sorting the nested tuple by track code (index 1)
sorted_tracks = sorted(track_tuple, key=lambda x: x[1])

# Display the sorted result
print("Sorted Tracks (by track code):", sorted_tracks)
```

```
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/IT-03/Desktop/S123/2B.py =====
Sorted Tracks (by track code): (('Track A', 20), ('Track B', 21), ('Track C', 22))
```

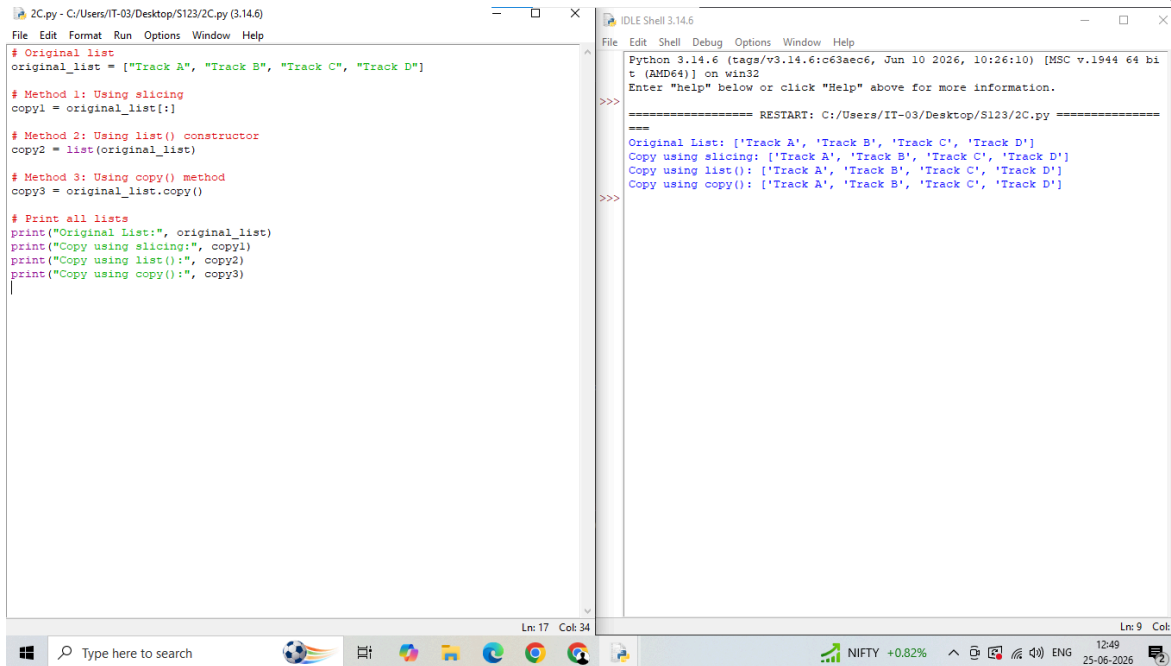
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Subject: Python for Data Science

Practical No. 2

2c. Write a python code to copy or clone list.



```
# 2C.py - C:/Users/IT-03/Desktop/S123/2C.py (3.14.6)
File Edit Format Run Options Window Help

# Original list
original_list = ["Track A", "Track B", "Track C", "Track D"]

# Method 1: Using slicing
copy1 = original_list[:]

# Method 2: Using list() constructor
copy2 = list(original_list)

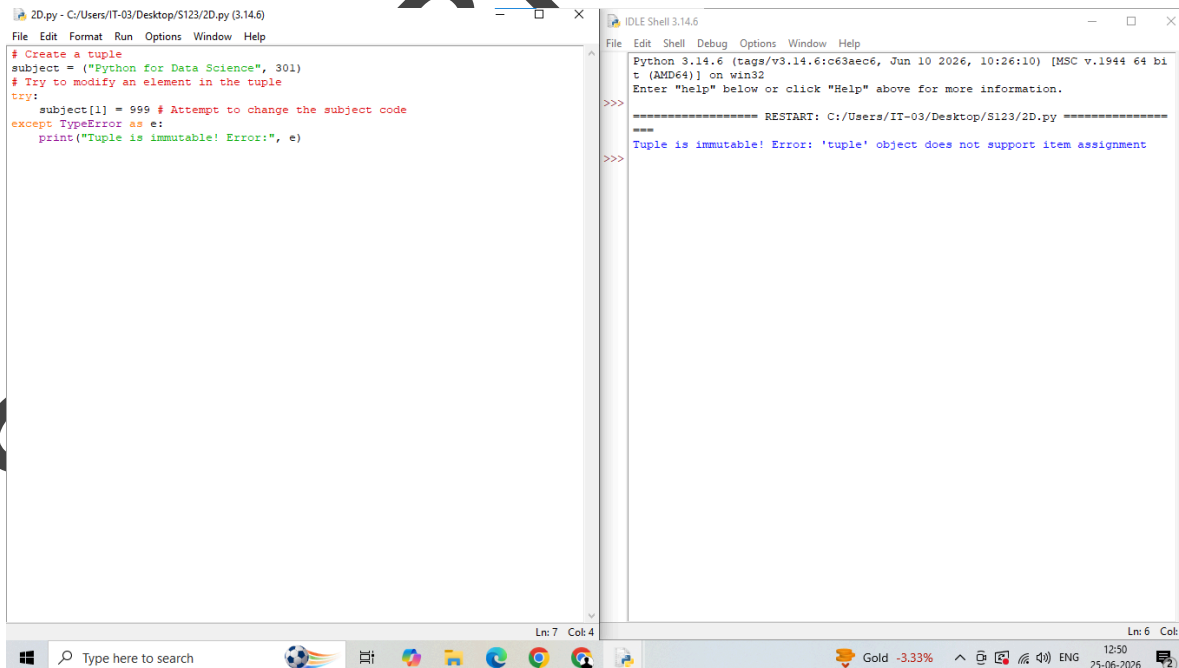
# Method 3: Using copy() method
copy3 = original_list.copy()

# Print all lists
print("Original List:", original_list)
print("Copy using slicing:", copy1)
print("Copy using list():", copy2)
print("Copy using copy():", copy3)
```

```
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bi
t (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/IT-03/Desktop/S123/2C.py =====
Original List: ['Track A', 'Track B', 'Track C', 'Track D']
Copy using slicing: ['Track A', 'Track B', 'Track C', 'Track D']
Copy using list(): ['Track A', 'Track B', 'Track C', 'Track D']
Copy using copy(): ['Track A', 'Track B', 'Track C', 'Track D']
>>>
```

2d. Write a python code to check immutability property of python tuples.



```
# 2D.py - C:/Users/IT-03/Desktop/S123/2D.py (3.14.6)
File Edit Format Run Options Window Help

# Create a tuple
subject = ("Python for Data Science", 301)
# Try to modify an element in the tuple
try:
    subject[1] = 999 # Attempt to change the subject code
except TypeError as e:
    print("Tuple is immutable! Error:", e)
```

```
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bi
t (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/IT-03/Desktop/S123/2D.py =====
Tuple is immutable! Error: 'tuple' object does not support item assignment
>>>
```

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Subject: Python for Data Science

Practical No. 2

Question 1: [Implement the below given questions in a single python file]

- Write a Python program to create a tuple with 5 different elements and print it.
- Write a program to access the first and last elements of a tuple using indexing.
- Write a program to slice a tuple and print the middle 3 elements.
- Write a program to concatenate two tuples and print the result.
- Write a program to reverse a tuple using slicing.
- Write a program to count how many times an element appears in a tuple.
- Write a program to find the index of a specific element in a tuple.
- Write a program to check if an element exists in a tuple.
- Write a program to convert a list to a tuple.
- Write a program to sort a tuple of numbers in ascending order.
- Write a program to repeat a tuple 3 times using the * operator.
- Write a program to check the immutability property of tuples by trying to modify an element.

SOL':

```
Q1.py - C:/Users/IT-03/Desktop/S123/Q1.py (3.14.6)
File Edit Format Run Options Window Help
# Question 1: Tuple Programs (a to l)

# a. Create a tuple with 5 different elements
my_tuple = ("Track A", "Track B", "Track C", "Track D", "Track E")
print("Original Tuple:", my_tuple)

# b. Access the first and last elements
print("\nFirst Element:", my_tuple[0])
print("Last Element:", my_tuple[-1])

# c. Slice the tuple and print the middle 3 elements
print("\nMiddle 3 Elements:", my_tuple[1:4])

# d. Concatenate two tuples
tuple2 = ("Track F", "Track G")
new_tuple = my_tuple + tuple2
print("\nConcatenated Tuple:", new_tuple)

# e. Reverse a tuple using slicing
print("\nReversed Tuple:", my_tuple[::-1])

# f. Count how many times an element appears
count_tuple = ("Track A", "Track B", "Track A", "Track C", "Track A")
print("\nCount of 'Track A':", count_tuple.count("Track A"))

# g. Find the index of a specific element
print("\nIndex of 'Track C':", my_tuple.index("Track C"))

# h. Check if an element exists in a tuple
element = "Track D"
if element in my_tuple:
    print("\n", element, "exists in the tuple.")
else:
    print("\n", element, "does not exist in the tuple.")

# i. Convert a list to a tuple
my_list = ["Track X", "Track Y", "Track Z"]
tuple_from_list = tuple(my_list)
print("\nTuple from List:", tuple_from_list)

# j. Sort a tuple of numbers in ascending order
tuple_of_numbers = (7, 12, 23, 45, 89)
sorted_tuple = sorted(tuple_of_numbers)
print("\nSorted Tuple:", sorted_tuple)

# k. Repeat a tuple 3 times using the * operator
repeated_tuple = ("Track A", "Track B", "Track C", "Track D", "Track E") * 3
print("\nRepeated Tuple:", repeated_tuple)

# l. Check the immutability property of tuples
tuple_obj = ("Track A", "Track B", "Track C")
try:
    tuple_obj[0] = "Track X"
except TypeError:
    print("\nTuple Immutability Test: 'tuple' object does not support item assignment")
```

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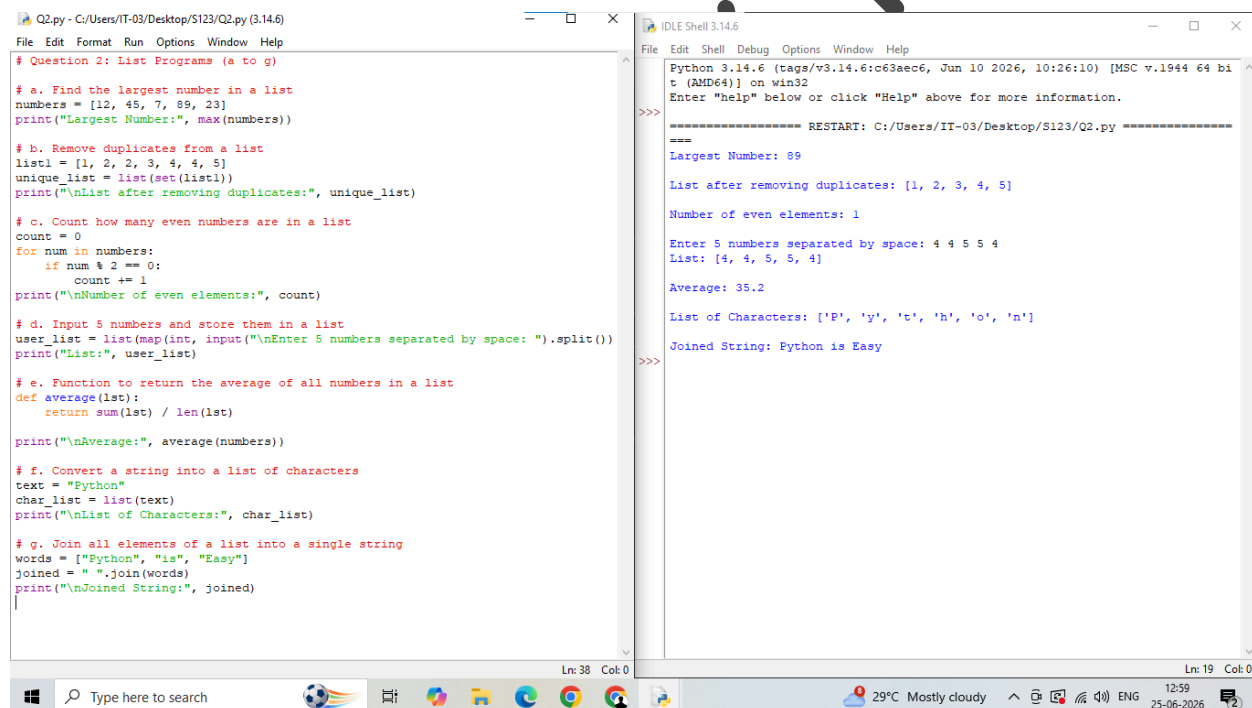
Subject: Python for Data Science

Practical No. 2

Question 2: [Implement the below given questions in a single python file]

- Write a program to find the largest number in a list.
- Write a program to remove duplicates from a list.
- Write a program to count how many even numbers are in a list.
- Write a program to input 5 numbers and store them in a list. Then display the list.
- Write a function that returns the average of all numbers in a list.
- Convert a string into a list of characters using list().
- Join all elements of a list into a single string using join().

SOL':



The screenshot shows a Python IDE with two windows. The left window displays the code for Question 2, and the right window shows the output of the program.

```
# Question 2: List Programs (a to g)

# a. Find the largest number in a list
numbers = [12, 45, 7, 89, 23]
print("Largest Number:", max(numbers))

# b. Remove duplicates from a list
list1 = [1, 2, 2, 3, 4, 4, 5]
unique_list = list(set(list1))
print("\nList after removing duplicates:", unique_list)

# c. Count how many even numbers are in a list
count = 0
for num in numbers:
    if num % 2 == 0:
        count += 1
print("\nNumber of even elements:", count)

# d. Input 5 numbers and store them in a list
user_list = list(map(int, input("\nEnter 5 numbers separated by space: ").split()))
print("List:", user_list)

# e. Function to return the average of all numbers in a list
def average(lst):
    return sum(lst) / len(lst)

print("\nAverage:", average(numbers))

# f. Convert a string into a list of characters
text = "Python"
char_list = list(text)
print("\nList of Characters:", char_list)

# g. Join all elements of a list into a single string
words = ["Python", "is", "Easy"]
joined = " ".join(words)
print("\nJoined String:", joined)
```

The output window shows the following results:

```
Python 3.14.6 (tags/v3.14.6:63a6ec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bi
t (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/IT-03/Desktop/S123/Q2.py =====
Largest Number: 89
List after removing duplicates: [1, 2, 3, 4, 5]
Number of even elements: 1
Enter 5 numbers separated by space: 4 4 5 5 4
List: [4, 4, 5, 5, 4]
Average: 35.2
List of Characters: ['P', 'y', 't', 'h', 'o', 'n']
Joined String: Python is Easy
>>>
```

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